



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Group Project

CPU & Hardware Performance Benchmarks

Computer Organisation & Hardware SECR1033

Lecturer : Dr. Mohd Foad Rohani

Section : 03

NAME	MATRIC NUMBER
YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201
SYARIFAH DANIA BINTI SYEDA ABU BAKAR	A23CS0183
AIN NURNABILA BINTI MOHD AZHAR	A23CS0207
ANIS SAFIYYA BINTI JANAI	A23CS0049

Video link: <https://youtu.be/GMBIGs7TMnI>

Table of Contents

1.0 Introduction.....	1
1.1 Problem Background	1
1.2 Aims.....	1
1.3 Objectives	2
1. 4 Scopes (Computer specification & benchmarking tools feature).....	2
2.0 Flowchart of the project	4
2.1 Flowchart of Project Part 1	4
2.2 Flowchart of Project Part 2	5
3.0 Benchmarking results	6
3.1 Benchmarking results using CPUZ.....	6
3.2 Benchmarking results using Geekbench	7
3.3 Benchmarking results using MiniTools	11
3.4 Benchmarking results using Windows Task Manager.....	13
4.0 Analysis, Comparison & Discussion	13
4.1 Analysis & discussion on benchmarking tools CPUZ	14
4.2 Analysis & discussion on benchmarking tools Geekbench	15
4.3 Analysis & discussion on benchmarking tools MiniTools	16
4.4 Analysis & discussion on benchmarking tools Task Manager	18
4.5 Analysis & discussion on Project Part 2	20
5.0 Conclusion, Reflection, and Reference.....	23
5.1 Conclusion	23
5.2 Reflection.....	23
Reference	24
Appendix.....	24

1.0 Introduction

A benchmark in computing is the process of executing a computer program, a collection of programs, or other processes to evaluate the relative performance of an item, usually by the administration of several common tests and trials.

A set of tests intended to gauge a computer or device's CPU performance is called a CPU benchmark (also known as CPU benchmarking). When comparing the performance of various systems under identical conditions, a set of standards, or baseline measures, are employed.

Software for CPU benchmarking is freely downloadable and available. We conducted the CPU benchmarking on three different computers in order to investigate and gain insights into computer performance, hardware differences, and the intricacies of software optimization.

1.1 Problem Background

Our central question revolves around how different CPUs perform under varying workloads. We aim to address the following key aspects:

- **Benchmarking CPUs:** We will select benchmarking tools and rigorously test different CPUs across various configurations. This process involves measuring execution times, memory usage, and other relevant metrics.
- **Hardware Impact:** By comparing CPUs, we'll gain insights into how hardware variations affect performance. Factors such as clock speed, cache size, and core count play a crucial role.
- **Software Optimization:** We'll explore the impact of software optimization on CPU workload handling.
- **Practical Application:** After conducting the benchmark on three different computers, we'll apply our findings practically by writing an assembly language program to benchmark a polynomial equation. This hands-on experience will deepen our understanding of CPU performance metrics.

1.2 Aims

- To compare the benchmarking results on three different computers using different benchmarking tools
- To run a simple assembly program on three different computers to calculate the difference in CPU running time between all three computers.

1.3 Objectives

- To recognize the fundamentals of CPU benchmarking by using CPU-Z, Geekbench, Windows Task Manager, and MiniTool Partition Wizard
- To differentiate each computer CPU performance if using every tools.
- To analyse how different applications could influence the performance of the CPU.
- To learn how to evaluate each benchmarking results by using the free tools.

1. 4 Scopes (Computer specification & benchmarking tools feature)

For this project, there are three computers that we are using. Table 1.4.1 shows each of the computer type and specification.

Specifications	Computer type		
	A	B	C
	Inspiron 14 5430	VivoBook X515EA A1500EA	HP ProBook 440 G2
Processor Type	Intel Core i7-1360P	Intel Core i5-1135G7	Intel Core i5-4210U
RAM Size	16 GB LPDDR5	8GB DDR4	8GB DDR3
GPU Type	Intel ® Iris ® Xe Graphics	Intel Iris Xe Graphics	Intel® HD Graphics Family
Motherboard	DELL Inc. 0GMW80 A00	ASUSTek COMPUTER INC.	Hewlett-Packard
Bus Specs	PCI-E Gen 4.0 (4x)	PCI-X	PCI-Express 2.0

Table 1.4.1

For this project, there are four distinct tools used which are CPU-Z, Geekbench, Task Manager, and MiniTool Partition Wizard.

1. CPU-Z

CPU-Z is a free software tool designed for system profiling and monitoring the performance of various computer components. Some of the information that can be obtained by using CPU-Z are:

- Processor specification
- Clock frequency or speed of the processor (CPU)
- Cache Levels
- Motherboard and bus specification
- Memory type and size

2. Geekbench

Geekbench is a cross-platform tool for comparing computer systems' performance. It offers a wide range of benchmarks to assess CPU, GPU, and memory performance. This benchmark generates scores that differentiate between single-core and multi-core performance by simulating real-world usage scenarios. The findings of the Geekbench Browser open platform to display the performance score. From Geekbench, we got:

- System Specifications
- CPU Test which consists of:
 - Single-Core Performance
 - Multi-Core Performance

3. Task Manager

Task Manager main function is to assist administrators and end users in monitoring, managing, and troubleshooting tasks. Within Task Manager, an application, a Windows process, or a background process can all be considered tasks. Task Manager offers details on hardware resource consumption and performance in relation to the applications and processes operating on the system. Also, administrators can identify unexpected or unusual behaviour that might point to malware or other illegal applications by using Task Manager.

4. MiniTool Partition Wizard

A disk management application called MiniTool Partition Wizard has a benchmarking feature that allows you to measure the read and write performance of storage devices. Some of the functions are:

- To run a benchmark against any disk
- Disk Benchmarking Process
- Stress Test Observations

2.0 Flowchart of the project

Flowchart of the project is important as it roughly show how the flow of the project will be conducted step by step.

2.1 Flowchart of Project Part 1

Figure 2.1 is the flowchart of project part 1. This is the flow of conducting the benchmarking comparison on 3 different computers. The benchmarking tools that we are using are CPU-Z, Geekbench, Windows Task Manager, and MiniTool Partition Wizard.

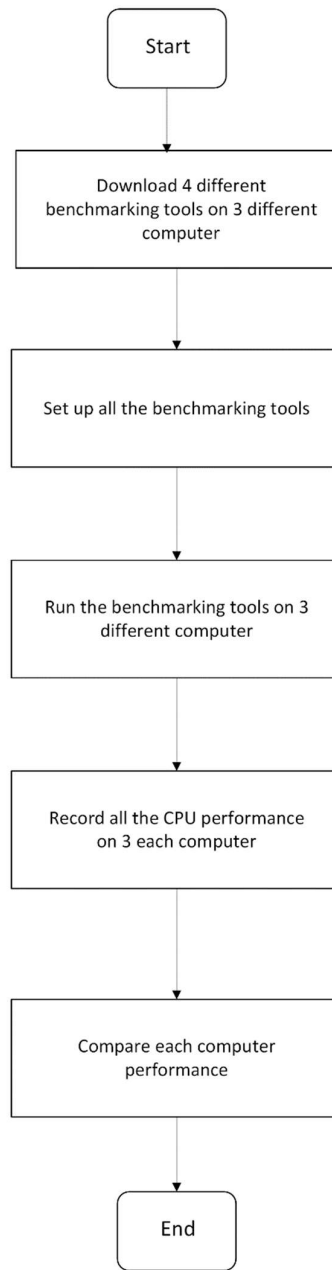


Figure 2.1

2.2 Flowchart of Project Part 2

Figure 2.2 is the flowchart of project part 2. This is the flow of calculating and measuring the time taken to execute different loops for each polynomial equation.

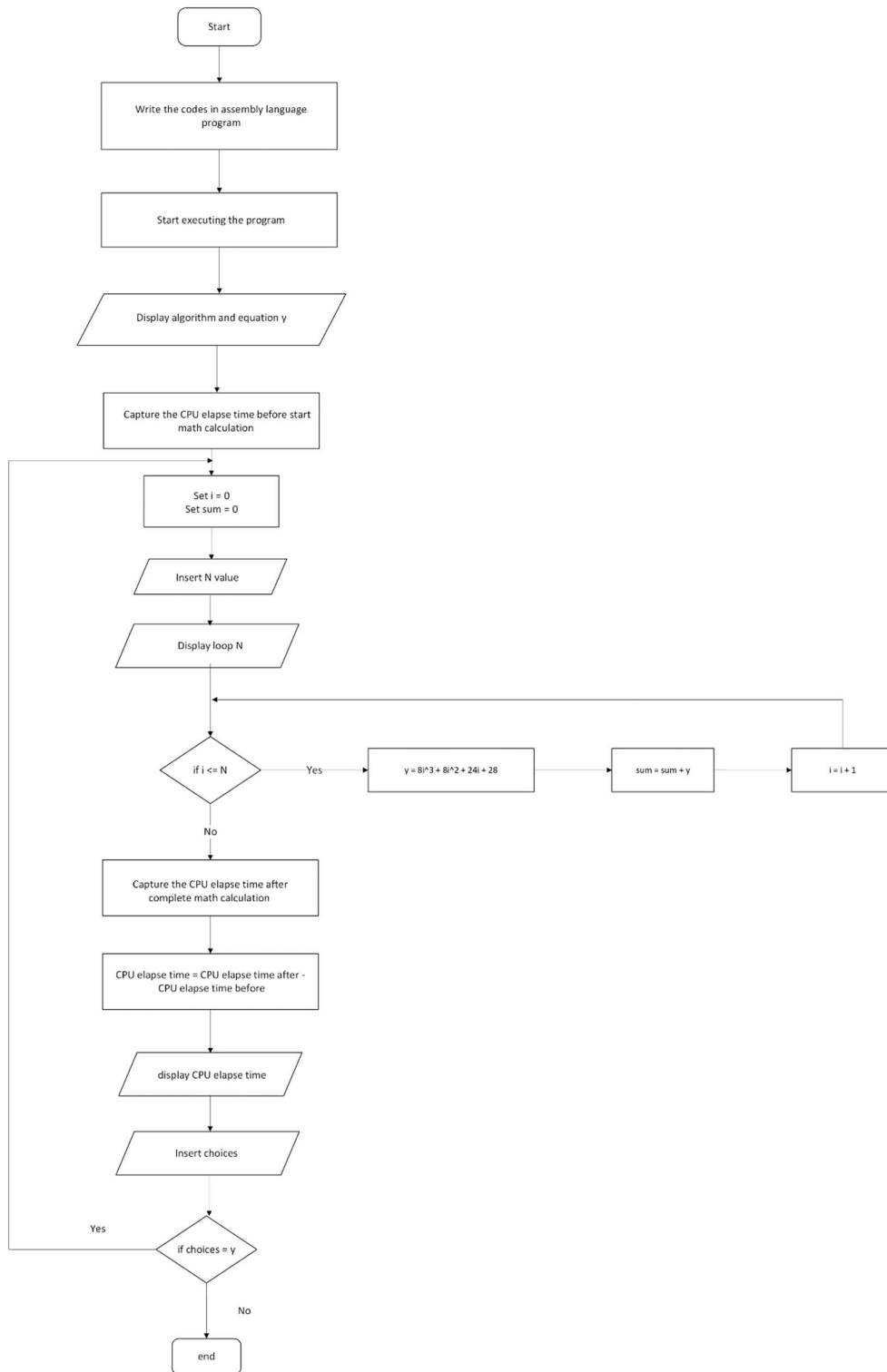


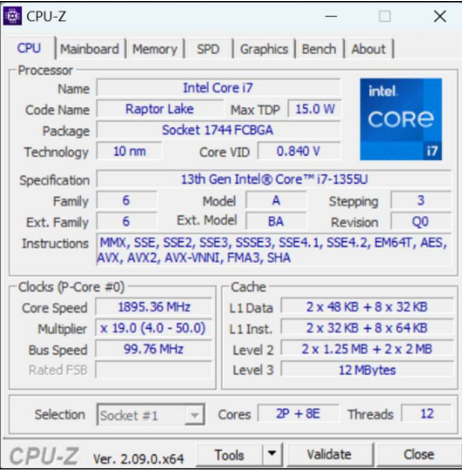
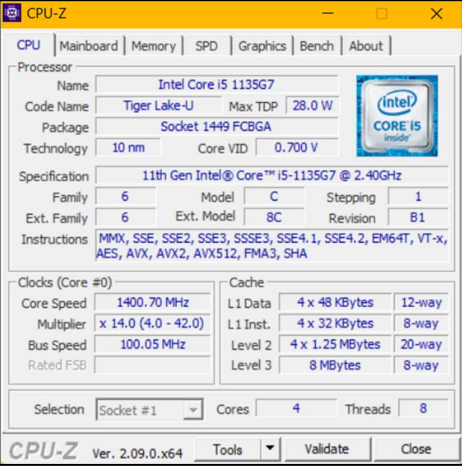
Figure 2.2

3.0 Benchmarking results

After conducting the benchmarking test on 3 different computers, below are the results of each benchmarking tools that we are using.

3.1 Benchmarking results using CPUZ

Table 3.1.1 is the result of benchmarking on 3 different computers using CPUZ benchmarking tools.

Computer	CPUZ (System Info)
A	
B	

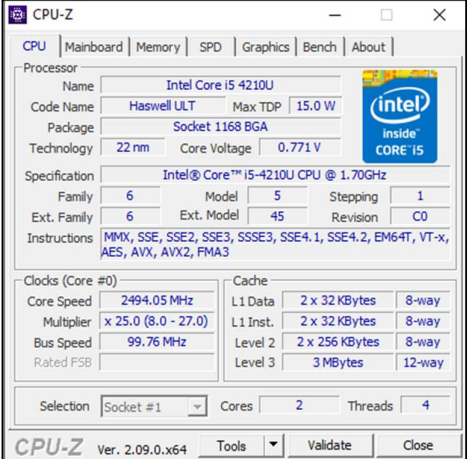
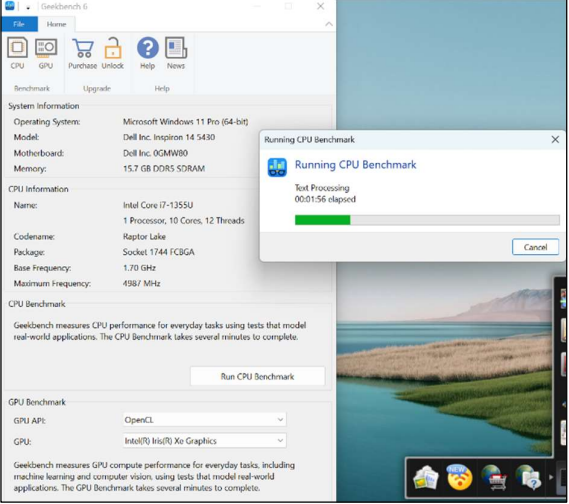
C	
---	--

Table 3.1.1

3.2 Benchmarking results using Geekbench

Table 3.2.1 is the result of benchmarking on 3 different computers using Geekbench benchmarking tools.

Computer	Geekbench (CPU stress test)
A	 CPU Information Name Intel Core i7-1355U Topology 1 Processor, 10 Cores, 12 Threads Identifier GenuineIntel Family 6 Model 186 Stepping 3 Base Frequency 1.70 GHz Cluster 1 2 Cores Cluster 2 8 Cores Maximum Frequency 4987 MHz Package Socket 1744 FCBGA

		Codename Raptor Lake L1 Instruction Cache 32.0 KB x 6 L1 Data Cache 48.0 KB x 6 L2 Cache 1.25 MB x 1 L3 Cache 12.0 MB x 1																																																																																																																																					
		<table> <tr> <th>Single-Core Score</th><th>928</th><th>Multi-Core Score</th><th>4289</th></tr> <tr> <td>File Compression</td><td>857</td><td>File Compression</td><td>3284</td></tr> <tr> <td></td><td>123.1 MB/sec</td><td></td><td>471.6 MB/sec</td></tr> <tr> <td>Navigation</td><td>669</td><td>Navigation</td><td>4319</td></tr> <tr> <td></td><td>4.03 routes/sec</td><td></td><td>26.0 routes/sec</td></tr> <tr> <td>HTML5 Browser</td><td>926</td><td>HTML5 Browser</td><td>4737</td></tr> <tr> <td></td><td>19.0 pages/sec</td><td></td><td>97.0 pages/sec</td></tr> <tr> <td>PDF Renderer</td><td>914</td><td>PDF Renderer</td><td>5066</td></tr> <tr> <td></td><td>21.1 Mpixels/sec</td><td></td><td>116.8 Mpixels/sec</td></tr> <tr> <td>Photo Library</td><td>1114</td><td>Photo Library</td><td>4726</td></tr> <tr> <td></td><td>15.1 images/sec</td><td></td><td>64.1 images/sec</td></tr> <tr> <td>Clang</td><td>888</td><td>Clang</td><td>5777</td></tr> <tr> <td></td><td>4.37 Klines/sec</td><td></td><td>28.5 Klines/sec</td></tr> <tr> <td>Text Processing</td><td>946</td><td>Text Processing</td><td>1130</td></tr> <tr> <td></td><td>75.8 pages/sec</td><td></td><td>90.5 pages/sec</td></tr> <tr> <td>Asset Compression</td><td>964</td><td>Asset Compression</td><td>6519</td></tr> <tr> <td></td><td>29.9 MB/sec</td><td></td><td>202.0 MB/sec</td></tr> <tr> <td>Object Detection</td><td>1052</td><td>Object Detection</td><td>2647</td></tr> <tr> <td></td><td>31.5 images/sec</td><td></td><td>79.2 images/sec</td></tr> <tr> <td>Background Blur</td><td>1275</td><td>Background Blur</td><td>4884</td></tr> <tr> <td></td><td>5.28 images/sec</td><td></td><td>20.2 images/sec</td></tr> <tr> <td>Horizon Detection</td><td>1067</td><td>Horizon Detection</td><td>6828</td></tr> <tr> <td></td><td>33.2 Mpixels/sec</td><td></td><td>212.5 Mpixels/sec</td></tr> <tr> <td>Object Remover</td><td>565</td><td>Object Remover</td><td>3909</td></tr> <tr> <td></td><td>43.4 Mpixels/sec</td><td></td><td>300.5 Mpixels/sec</td></tr> <tr> <td>HDR</td><td>895</td><td>HDR</td><td>4477</td></tr> <tr> <td></td><td>26.3 Mpixels/sec</td><td></td><td>131.4 Mpixels/sec</td></tr> <tr> <td>Photo Filter</td><td>1062</td><td>Photo Filter</td><td>4944</td></tr> <tr> <td></td><td>10.5 images/sec</td><td></td><td>49.1 images/sec</td></tr> <tr> <td>Ray Tracer</td><td>1039</td><td>Ray Tracer</td><td>6666</td></tr> <tr> <td></td><td>1.01 Mpixels/sec</td><td></td><td>6.45 Mpixels/sec</td></tr> <tr> <td>Structure from Motion</td><td>922</td><td>Structure from Motion</td><td>5866</td></tr> <tr> <td></td><td>29.2 Kpixels/sec</td><td></td><td>185.7 Kpixels/sec</td></tr> </table>	Single-Core Score	928	Multi-Core Score	4289	File Compression	857	File Compression	3284		123.1 MB/sec		471.6 MB/sec	Navigation	669	Navigation	4319		4.03 routes/sec		26.0 routes/sec	HTML5 Browser	926	HTML5 Browser	4737		19.0 pages/sec		97.0 pages/sec	PDF Renderer	914	PDF Renderer	5066		21.1 Mpixels/sec		116.8 Mpixels/sec	Photo Library	1114	Photo Library	4726		15.1 images/sec		64.1 images/sec	Clang	888	Clang	5777		4.37 Klines/sec		28.5 Klines/sec	Text Processing	946	Text Processing	1130		75.8 pages/sec		90.5 pages/sec	Asset Compression	964	Asset Compression	6519		29.9 MB/sec		202.0 MB/sec	Object Detection	1052	Object Detection	2647		31.5 images/sec		79.2 images/sec	Background Blur	1275	Background Blur	4884		5.28 images/sec		20.2 images/sec	Horizon Detection	1067	Horizon Detection	6828		33.2 Mpixels/sec		212.5 Mpixels/sec	Object Remover	565	Object Remover	3909		43.4 Mpixels/sec		300.5 Mpixels/sec	HDR	895	HDR	4477		26.3 Mpixels/sec		131.4 Mpixels/sec	Photo Filter	1062	Photo Filter	4944		10.5 images/sec		49.1 images/sec	Ray Tracer	1039	Ray Tracer	6666		1.01 Mpixels/sec		6.45 Mpixels/sec	Structure from Motion	922	Structure from Motion	5866		29.2 Kpixels/sec		185.7 Kpixels/sec	
Single-Core Score	928	Multi-Core Score	4289																																																																																																																																				
File Compression	857	File Compression	3284																																																																																																																																				
	123.1 MB/sec		471.6 MB/sec																																																																																																																																				
Navigation	669	Navigation	4319																																																																																																																																				
	4.03 routes/sec		26.0 routes/sec																																																																																																																																				
HTML5 Browser	926	HTML5 Browser	4737																																																																																																																																				
	19.0 pages/sec		97.0 pages/sec																																																																																																																																				
PDF Renderer	914	PDF Renderer	5066																																																																																																																																				
	21.1 Mpixels/sec		116.8 Mpixels/sec																																																																																																																																				
Photo Library	1114	Photo Library	4726																																																																																																																																				
	15.1 images/sec		64.1 images/sec																																																																																																																																				
Clang	888	Clang	5777																																																																																																																																				
	4.37 Klines/sec		28.5 Klines/sec																																																																																																																																				
Text Processing	946	Text Processing	1130																																																																																																																																				
	75.8 pages/sec		90.5 pages/sec																																																																																																																																				
Asset Compression	964	Asset Compression	6519																																																																																																																																				
	29.9 MB/sec		202.0 MB/sec																																																																																																																																				
Object Detection	1052	Object Detection	2647																																																																																																																																				
	31.5 images/sec		79.2 images/sec																																																																																																																																				
Background Blur	1275	Background Blur	4884																																																																																																																																				
	5.28 images/sec		20.2 images/sec																																																																																																																																				
Horizon Detection	1067	Horizon Detection	6828																																																																																																																																				
	33.2 Mpixels/sec		212.5 Mpixels/sec																																																																																																																																				
Object Remover	565	Object Remover	3909																																																																																																																																				
	43.4 Mpixels/sec		300.5 Mpixels/sec																																																																																																																																				
HDR	895	HDR	4477																																																																																																																																				
	26.3 Mpixels/sec		131.4 Mpixels/sec																																																																																																																																				
Photo Filter	1062	Photo Filter	4944																																																																																																																																				
	10.5 images/sec		49.1 images/sec																																																																																																																																				
Ray Tracer	1039	Ray Tracer	6666																																																																																																																																				
	1.01 Mpixels/sec		6.45 Mpixels/sec																																																																																																																																				
Structure from Motion	922	Structure from Motion	5866																																																																																																																																				
	29.2 Kpixels/sec		185.7 Kpixels/sec																																																																																																																																				

B

CPU Information

Name

Intel Core i5-1135G7

Topology

1 Processor, 4 Cores, 8 Threads

Identifier

GenuineIntel Family 6 Model 140 Stepping 1

Base Frequency

2.39 GHz

Cluster 1

4 Cores

Maximum Frequency

4186 MHz

Package

Socket 1449 FCBGA

Codename

Tiger Lake-U

L1 Instruction Cache

32.0 KB x 4

L1 Data Cache

48.0 KB x 4

L2 Cache

1.25 MB x 4

L3 Cache

8.00 MB x 1

Single-Core Performance

Single-Core Score

1382

File Compression

1310

188.2 MB/sec

Navigation

1330

8.01 routes/sec

HTML5 Browser

1289

26.4 pages/sec

PDF Renderer

1371

31.6 Mpixels/sec

Photo Library

1379

18.7 images/sec

Clang

1378

6.79 Klines/sec

Text Processing

1332

106.7 pages/sec

Asset Compression

1430

44.3 MB/sec

Object Detection

1405

42.0 images/sec

Background Blur

1659

6.87 images/sec

Horizon Detection

1886

58.7 Mpixels/sec

Object Remover

1052

80.9 Mpixels/sec

HDR

1336

39.2 Mpixels/sec

Photo Filter

1648

16.4 images/sec

Ray Tracer

1098

1.06 Mpixels/sec

Structure from Motion

1518

48.1 Kpixels/sec

Multi-Core Performance

Multi-Core Score

3495

File Compression

1733

248.8 MB/sec

Navigation

4644

28.0 routes/sec

HTML5 Browser

3085

63.2 pages/sec

PDF Renderer

4211

97.1 Mpixels/sec

Photo Library

4439

60.2 images/sec

Clang

5182

25.5 Klines/sec

Text Processing

1629

130.5 pages/sec

Asset Compression

5726

177.4 MB/sec

Object Detection

2012

60.2 images/sec

Background Blur

4273

17.7 images/sec

Horizon Detection

4656

144.9 Mpixels/sec

Object Remover

3390

260.6 Mpixels/sec

HDR

3690

108.3 Mpixels/sec

Photo Filter

2732

27.1 images/sec

Ray Tracer

4953

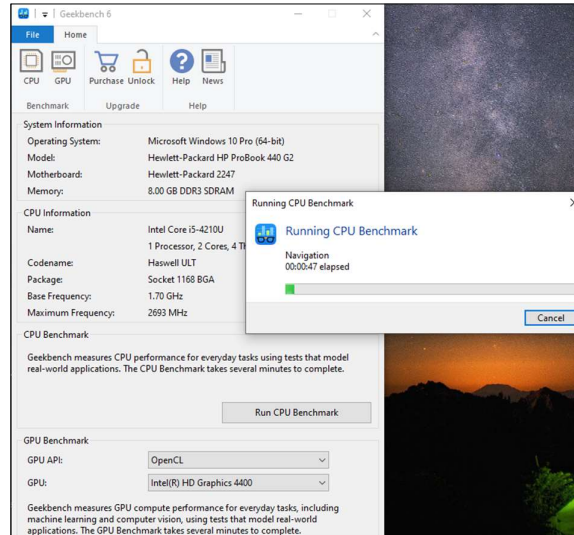
4.79 Mpixels/sec

Structure from Motion

4159

131.7 Kpixels/sec

C



CPU Information	
Topology	1 Processor, 2 Cores, 4 Threads
Identifier	GenuineIntel Family 6 Model 69 Stepping 1
Base Frequency	1.70 GHz
Cluster 1	2 Cores
Maximum Frequency	2693 MHz
Package	Socket 1168 BGA
Codename	Haswell ULT
L1 Instruction Cache	32.0 KB x 2
L1 Data Cache	32.0 KB x 2
L2 Cache	256 KB x 2
L3 Cache	3.00 MB x 1

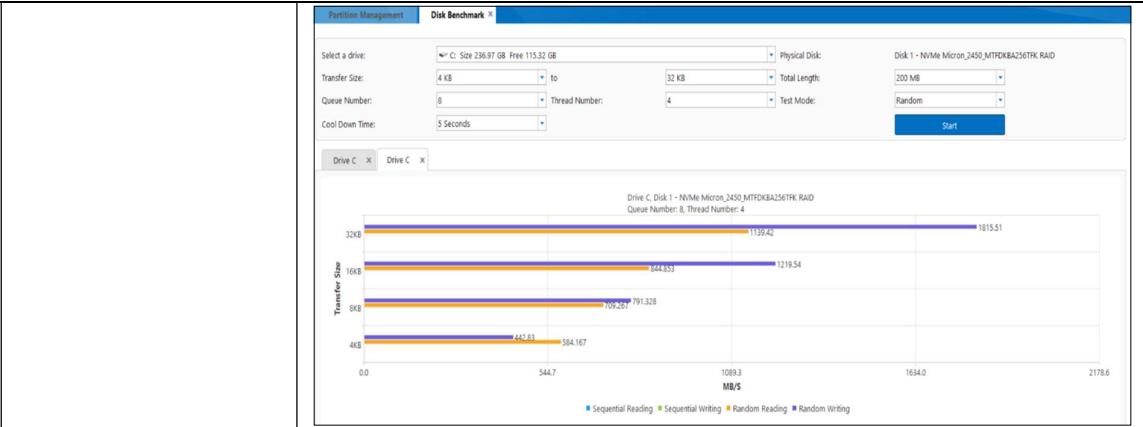
Single-Core Score	487	Multi-Core Score	1043
File Compression	444	File Compression	734
Navigation	63.8 MB/sec	Navigation	105.4 MB/sec
HTML5 Browser	563	Navigation	1521
PDF Renderer	3.39 routes/sec	HTML5 Browser	1229
Photo Library	555	PDF Renderer	25.2 pages/sec
Clang	11.4 pages/sec	PDF Renderer	1276
Text Processing	501	Photo Library	29.4 Mpixels/sec
Asset Compression	11.6 Mpixels/sec	Photo Library	958
Object Detection	435	Clang	13.0 images/sec
Background Blur	5.90 images/sec	Clang	1208
Horizon Detection	536	Text Processing	5.95 Klines/sec
Object Remover	2.64 Klines/sec	Text Processing	544
HDR	455	Asset Compression	43.5 pages/sec
Photo Filter	532	Asset Compression	1250
Ray Tracer	16.5 MB/sec	Object Detection	38.7 MB/sec
Structure from Motion	205	Object Detection	442
	6.15 images/sec	Background Blur	13.2 images/sec
	685	Background Blur	1305
	2.84 images/sec	Horizon Detection	5.40 images/sec
	757	Horizon Detection	1595
	23.6 Mpixels/sec	Object Remover	49.6 Mpixels/sec
	457	Object Remover	933
	35.1 Mpixels/sec	HDR	71.7 Mpixels/sec
	545	HDR	1135
	16.0 Mpixels/sec	Photo Filter	33.3 Mpixels/sec
	589	Photo Filter	1272
	5.84 images/sec	Ray Tracer	12.6 images/sec
	384	Ray Tracer	1368
	371.3 Kpixels/sec	Structure from Motion	1.32 Mpixels/sec
	561	Structure from Motion	1239
	17.7 Kpixels/sec		39.2 Kpixels/sec

Table 3.2.1

3.3 Benchmarking results using MiniTools

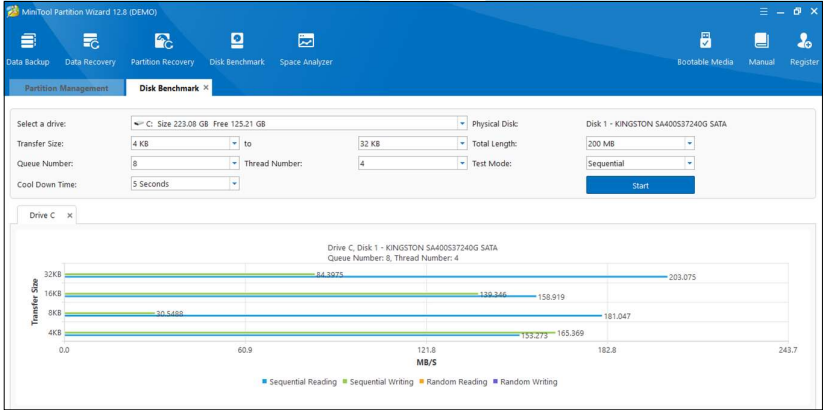
Table 3.3.1 is the result of benchmarking on 3 different computers using MiniTools Partition Wizard Software benchmarking tools.

Computer	MiniTools (Disk benchmark)															
A	Test mode: Sequential <table><tr><th>Transfer Size</th><th>Sequential Reading (MB/s)</th><th>Sequential Writing (MB/s)</th></tr><tr><td>32KB</td><td>2438.18</td><td>1740.88</td></tr><tr><td>16KB</td><td>1364.19</td><td>1364.19</td></tr><tr><td>8KB</td><td>1106.04</td><td>602.062</td></tr><tr><td>4KB</td><td>602.062</td><td>602.062</td></tr></table>	Transfer Size	Sequential Reading (MB/s)	Sequential Writing (MB/s)	32KB	2438.18	1740.88	16KB	1364.19	1364.19	8KB	1106.04	602.062	4KB	602.062	602.062
Transfer Size	Sequential Reading (MB/s)	Sequential Writing (MB/s)														
32KB	2438.18	1740.88														
16KB	1364.19	1364.19														
8KB	1106.04	602.062														
4KB	602.062	602.062														
	Test mode: Random <table><tr><th>Transfer Size</th><th>Random Reading (MB/s)</th><th>Random Writing (MB/s)</th></tr><tr><td>32KB</td><td>2552.76</td><td>1275.84</td></tr><tr><td>16KB</td><td>1428.42</td><td>704.999</td></tr><tr><td>8KB</td><td>1185.99</td><td>352.576</td></tr><tr><td>4KB</td><td>623.292</td><td>623.292</td></tr></table>	Transfer Size	Random Reading (MB/s)	Random Writing (MB/s)	32KB	2552.76	1275.84	16KB	1428.42	704.999	8KB	1185.99	352.576	4KB	623.292	623.292
Transfer Size	Random Reading (MB/s)	Random Writing (MB/s)														
32KB	2552.76	1275.84														
16KB	1428.42	704.999														
8KB	1185.99	352.576														
4KB	623.292	623.292														
B	Test mode: Sequential <table><tr><th>Transfer Size</th><th>Sequential Reading (MB/s)</th><th>Sequential Writing (MB/s)</th></tr><tr><td>32KB</td><td>1916.31</td><td>1528.84</td></tr><tr><td>16KB</td><td>1226.57</td><td>1155.7</td></tr><tr><td>8KB</td><td>1226.57</td><td>778.828</td></tr><tr><td>4KB</td><td>578.082</td><td>578.082</td></tr></table>	Transfer Size	Sequential Reading (MB/s)	Sequential Writing (MB/s)	32KB	1916.31	1528.84	16KB	1226.57	1155.7	8KB	1226.57	778.828	4KB	578.082	578.082
Transfer Size	Sequential Reading (MB/s)	Sequential Writing (MB/s)														
32KB	1916.31	1528.84														
16KB	1226.57	1155.7														
8KB	1226.57	778.828														
4KB	578.082	578.082														
	Test mode: Random															



C

Test mode: Sequential



Test mode: Random

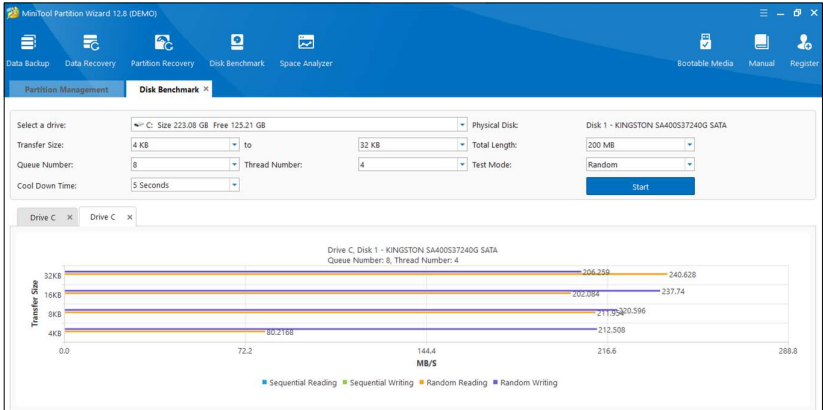


Table 3.3.1

3.4 Benchmarking results using Windows Task Manager

Table 3.4.1 is the result of benchmarking on 3 different computers using Task Manager benchmarking tools.

Com puter	Task Manager (Active Program and Resource Adjustment)											
	Low load				Middle load				High load			
A	Processes											
	Name	Status	CPU	Memory	Disk	Network	CPU	Memory	Disk	Network	CPU	Memory
B	Processes											
	Name	Status	CPU	Memory	Disk	Network	CPU	Memory	Disk	Network	CPU	Memory
C	Processes											
	Name	Status	CPU	Memory	Disk	Network	CPU	Memory	Disk	Network	CPU	Memory

Table3.4.1

4.0 Analysis, Comparison & Discussion

After conducting the benchmarking test on 3 different computer and gained the results, we would interpret the results and compare them to see the difference performance of each computer.

4.1 Analysis & discussion on benchmarking tools CPUZ

Table 4.1.1 are the discussion from analysing the results using benchmarking tools CPUZ. We compare the results for each computer and put it in the form of table.

Computer	CPUZ (System Info)
A	This computer is a 13th generation Intel Core i7 CPU from the Raptor Lake series runs at a base clock speed of about 1.9 GHz. It has six cores and twelve threads, with its cutting-edge instruction sets and technology, it can handle a variety of high-performance jobs. Multiple cache layers are included in the clearly specified cache structure to improve processing performance. The CPU has a low power consumption of 15W TDP and is produced with a 10 nm production technology.
B	Computer B is equipped with an Intel Core i5-1135G7 CPU from the Tiger Lake series. It features four cores and eight threads, allowing it to handle multiple tasks simultaneously. With four cores and eight threads, this CPU is well-suited for multitasking and running multi-threaded applications, enhancing productivity for users who run multiple applications simultaneously. The CPU is designed with a thermal design power (TDP) of 28W, making it energy-efficient while maintaining high performance, suitable for laptops that require a balance between power and battery life.
C	For computer C, the CPU processor is an Intel Core i5-4210U with a clock frequency of 1.70 GHz, which can boost up to around 2.49 GHz. This feature allows the processor to allocate more power to the CPU when it is handling demanding tasks, thereby enhancing performance when needed. The cache includes 2 x 32 KBytes of L1 Data cache, 2 x 32 KBytes of L1 Instruction cache, 2 x 256 KBytes of L2 cache, and a 3 MBytes L3 cache. The cache structure is designed to improve data processing efficiency. Next, the memory installed is 8 GB of DDR3 RAM. DDR3 RAM is an older generation of memory compared to the latest DDR4 or DDR5, but it is still sufficient for many computing tasks. This computer uses HP 2247 motherboard which supports PCI Express 2.0, allowing high-speed connections for components like graphics, network, and storage cards. It offers data transfer rates of 5.0 Giga transfers per second.

To compare each three computers based on CPU-Z benchmarking, computer A is the best overall in terms of performance and efficiency, followed by Computer B which provides a good balance for everyday tasks and multitasking. Computer C is suitable for basic usage but falls behind the other two in terms of technology and performance.

4.2 Analysis & discussion on benchmarking tools Geekbench

Table 4.2.1 are the discussion from analysing the results using benchmarking tools Geekbench. We compare the results for each computer and put it in the form of table.

Computer	Geekbench (CPU stress test)
A	Overall scores of Single-core Performance are 982. It suggests that it performs effectively in a task that only uses one thread. It indicates that tasks can only be performed sequentially and cannot be parallelized. This works well for a variety of common applications with low workloads. Nevertheless, the weaknesses in this Single-core Performance are the navigations and horizon detection. For example, when low navigation means that it has a slower map executing and loading in applications such as Google Maps. Overall scores of Multi-core Performance are 4289. It indicates that it performs effectively in multi-threaded applications. It means that the tasks can be split up into several parallel threads and handled simultaneously by various cores by the CPU. In simpler sentences, this can work great for multitasking and rendering tasks. But it also has some weaknesses such as background blur where the scores is 4884. It means that there will be little lag when handling photos or videos with a lot of background blur.
B	Computer B achieved the following scores in Geekbench: - Single-Core Performance: The single-core performance score is 1382. This indicates strong efficiency in tasks that require single-thread processing. It is suitable for applications such as web browsing, document editing, and other light to moderate tasks. - Multi-Core Performance: The multi-core performance score is 3495. This reflects the CPU's ability to handle multi-threaded applications efficiently. It performs well in tasks such as video encoding, software development, and other applications that can utilize multiple cores and threads effectively. These results suggest that Computer B is capable of managing both single-threaded and multi-threaded workloads efficiently. Its performance in Geekbench shows a good balance between single-core and multi-core capabilities, making it versatile for various computing tasks.
C	The Geekbench results show that it performs moderately well on tasks requiring both single and many cores. The 487 single-core score suggests sufficient performance for common tasks like word processing and navigation. However, it indicates that it might have trouble with tasks that mostly rely on single-threaded performance. Its low results in image filtering and photo library management suggest limitations when it comes to conducting graphics-intensive activities without a dedicated GPU. Its enhanced capacity to manage heavier workloads and multitasking is shown in its multi-core score of 1043, with significant improvements in tasks including PDF rendering and photo library management. With its dual-core, four-thread design and extensive cache structure, this performance profile validates the information from CPU-Z and displays the processor's efficiency and versatility for a variety of computer tasks.



Computer A excels in multi-core performance and is suitable for multitasking and rendering tasks, making it the top choice for high-performance needs. **Computer B** offers the best single-core performance and strong multi-core performance, making it versatile for a wide range of tasks. **Computer C** is the least powerful, sufficient for basic tasks but not suitable for demanding applications.

4.3 Analysis & discussion on benchmarking tools MiniTools

Table 4.3.1 are the discussion from analysing the results using benchmarking tools MiniTools. We compare the results for each computer and put it in the form of table.

Computer	MiniTools (Disk benchmark)
A	Sequential mode results demonstrate the speed at which the disk can read and write data when it is being accessed continuously. As we can see that as the transfer size increasing, the performance is also increasing. For example, at the largest transfer size (32 KB), it has speeds for sequential reading at 2128.05 MB/s meanwhile for sequential writing is at 2438.18 MB/s. This can conclude that computer A performs very well with higher transfer size, and it can work very efficiently in handling the continuous data stream. Random mode results demonstrate that speed at which the disk can read or write data in an undefined sequence without following any pattern. As we can see that transfer size increasing, the performance is also increasing. For example, at the largest transfer size (32 KB), it has speeds for sequential reading at 2149.59 MB/s meanwhile for sequential writing is at 2552.76 MB/s. We can conclude that computer A is quite efficient in managing the non-linear data access patterns that are frequently seen in daily computer use. Overall, computer A fast random read and write speeds particularly at larger transfer sizes show the capability to provide outstanding performance in a variety of settings, from daily use to demanding business tasks.

B	Sequential mode results demonstrate that as the transfer size increases, the performance does not follow a strictly linear pattern. For example, at the largest transfer size (32KB), it has speeds for sequential reading at 203.075 MB/s and for sequential writing at 84.3975 MB/s. This suggests that Computer B performs efficiently in handling continuous data streams, though the performance can vary depending on the specific transfer size. Random mode results show that as the transfer size increases, the performance generally improves, with some variability. For example, at the largest transfer size (32KB), it has speeds for random reading at 240.628 MB/s and for random writing at 206.259 MB/s. This indicates that Computer B is quite efficient in managing non-linear data access patterns that are frequently seen in daily computer use. Overall, Computer B demonstrates good performance in both sequential and random read and write operations across various transfer sizes. Its fast random read and write speeds, particularly at larger transfer sizes, show its capability to provide outstanding performance in a variety of settings, from daily use to more demanding business tasks.
C	For the Kingston SA400S37240G SATA SSD, the MiniTool Partition Wizard test offers an in-depth evaluation of the drive's performance in many scenarios. The SSD demonstrated great read and write speeds in sequential test mode, reaching up to 281.075 MB/s and 186.019 MB/s, respectively. Excellent efficiency is shown by these findings when it comes to tasks involving large, continuous data transfers, like installing huge software and backing up systems. The random test mode results, on the other hand, demonstrated read rates ranging from 72.2 MB/s to 240.628 MB/s and write rates ranging from 144.4 MB/s to 212.508 MB/s. These results were meant to imitate real-world usage scenarios when several small files are accessed simultaneously. Due to extra work involved in retrieving dispersed data across the disk, these speeds are often slower than sequential ones which could affect performance of the tasks.

Table 4.3.1

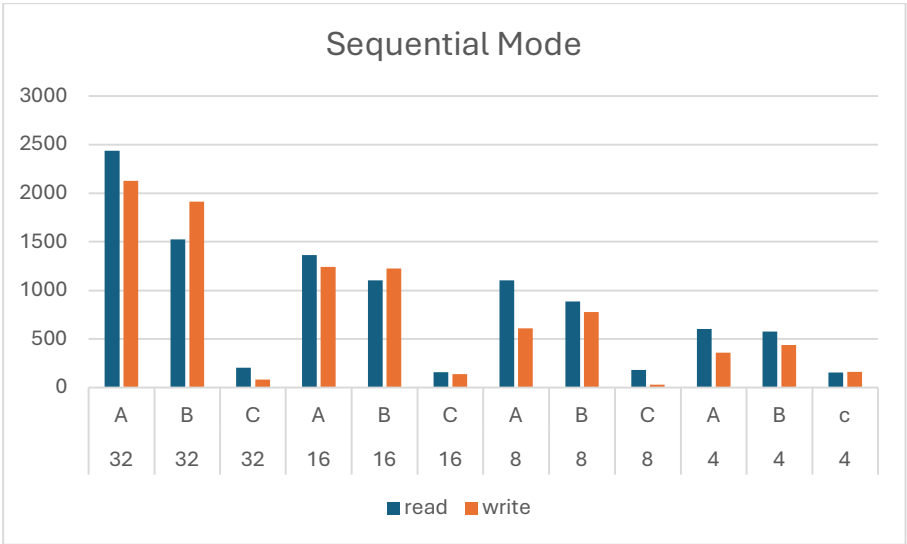


Figure 4.3.1



Figure 4.3.2

Computer A has an excellent sequential and random performance with incredible read and write speeds that make it ideal for managing massive, continuous data streams and non-linear data access patterns, though it may be excessive for an average user.

While Computer B has strong adaptability and stable performance throughout a range of transfer sizes, its variable performance and slower sequential write speed might be an issue in more demanding situations.

A wide range of tasks can be accomplished using Computer C's moderate sequential performance and random speeds. However, tasks needing high random access speeds may be affected by Computer C's less consistent and more unpredictable random performance.

4.4 Analysis & discussion on benchmarking tools Task Manager

Table 4.4.1 are the discussion from analysing the results using benchmarking tools Task Manager. We compare the results for each computer and put it in the form of table.

Computer	Task Manager (Active Program and Resource Adjustment)
A	From the results, it shows that 46% of the memory being used, which mean that the system is not under any stress, and it is in a low load condition. The system memory now is handling a low task where it only open the Microsoft Edge only. Relatively low CPU, disk, and network utilization points to little system activity. The analysis demonstrates that the system effectively manages resources with little impact from individual operations. At the 58% of the memory, it indicates that the system is operating under the medium load. Microsoft Edge continues to be a large memory consumer, then followed by the windows explorer and google chrome. It shows that, by running a lot of software at the same time, it could increase the percentage of the memory because it has to handle a lot of tasks simultaneously. At 76% of the memory, it indicate that the system is operating under the high load. Where, Microsoft Edge continues to be a large memory consumer followed by

	Windows Explorer and Desktop Window Manager. This software consumes a lot of RAM. But the Microsoft Visual Studio plays such a big role in increasing the memory percentage. This thorough analysis shows that while memory use is in a high range, the system is managing resources under high demand with higher CPU activity.
B	At low load, Computer B's memory usage is 50%. This indicates that the system is handling lightweight tasks efficiently without exerting significant strain on its memory resources. Typical applications running during low load scenarios include basic system processes, web browsing, document editing, and other light activities. The biggest memory consumption is McAfee . The memory usage is well within the system's capacity, ensuring smooth performance and responsiveness. Under medium load, the memory usage increases to 57%. This slight increase suggests that the system is managing a moderate level of activity, which may include running multiple applications simultaneously, such as web browsers with multiple tabs open, office productivity software, and perhaps some light multimedia tasks. The system's resource management remains effective, ensuring that applications run smoothly without any significant performance degradation. At high load, the memory usage rises to 83 %. This substantial increase indicates that the system is under significant stress, likely due to running demanding applications such as development tools, multimedia editing software, or having many applications and processes running concurrently. The high memory usage shows that Computer B is approaching its memory capacity limits, which could lead to potential slowdowns or the need for the system to use virtual memory (paging), which can impact performance.
C	The Task Manager indicates that memory is being used at 47% and CPU at just 2%. The system is using 0% of its disk, network, and GPU resources, which suggests that the tasks it is performing now require minimal hardware resources. The main programs, including Task Manager and Windows Explorer, likewise use few resources. their CPU and memory utilization are 0.8% and 0.1%, respectively. The background processes that were currently operating also consume very little resources which highlights the system's effective resource management and allocation under low demand circumstances. Performance is limited in its resource consumption under middle load levels. The Task Manager shows that memory use has increased to 53% and CPU usage is slightly higher at 7%. With very low usage of 1% or 0% for disk, network, and GPU, the system can do more tasks without causing a large impact on its hardware resources. The applications that were operating used a moderate amount of memory and CPU power. Most background programs continue to show minimal memory use and 0% CPU usage. All things considered, the system operates effectively under a middle load condition, using few resources and consuming little power. The high load is primarily due to high memory usage, which stands at 91% as indicated by the Task Manager. Reduced performance and system slowdowns can arise from this excessive memory usage. The Firefox browser, which uses over 2687.4 MB of RAM, is the main cause of this high memory utilization. Apart from Firefox, some other applications are also consuming a significant amount of memory, although to a lower extent. The system has less RAM available for other processes when numerous apps and services take up a significant amount of the memory, which could occasionally cause performance problems and make the system unavailable.

Table 4.4.1

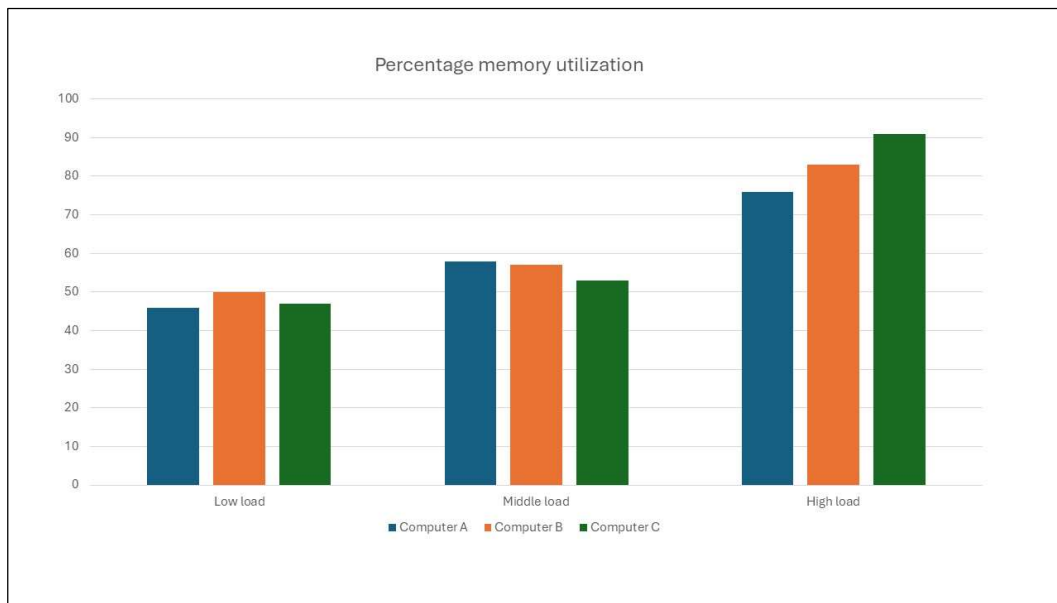


Figure 4.4.1

After comparing the 3 computers at the level of low load conditions, it shows that these 3 computers have efficient resource management, smooth functionality and efficiency during periods of low demand with little impact from any individual tasks.

Next, at the middle load, with increased tasks and moderate activity, the three computers show effective resource management under medium load applications too without obvious decreases in performance or unnecessary resource usage.

Lastly, when handling heavy load conditions, the three computers show different levels of efficiency. The high memory utilization suggests possible stress and performance issues, especially for Computers B and C, which are approaching or beyond their memory capacity limits, even though they are able to handle high resource demands. In these high-demand settings, maintaining performance still requires effective resource management.

4.5 Analysis & discussion on Project Part 2

Many factors could influence the CPU consumption time, which indicates a computer's performance directly. Operating system and program variations, memory and hardware variations, background activities, and system load are some of the elements that could impacted CPU running time.

We have run the assembly program on 3 different computers in order to observe the CPU elapse time for 3 computers. The program is about looping a mathematical equation for 50,100,200,500,700, and 1000 times by pressing start without debugging (F5) function. By calling the “GetMseconds” function, we could get the number of milliseconds that have been elapsed.

Based on the information that we gained from running the assembly program for 50 loops, the CPU elapse time for running the program on computer C was 4623 ms, while on computer B was 4745 ms, and on computer A was 4923 ms. By looking at each CPU elapse time for each computer, we can prove that computer C has a better performance compared to computer B and Computer A.

Next, based on the information that we gained from running the assembly program for 100 loops, the CPU elapse time for running the program on computer C was 9240 ms, while on computer B was 9477 ms, and on computer A was 9685 ms. By looking at each CPU elapse time for each computer, we can prove that computer C has a better performance compared to computer B and Computer A.

Next, based on the information that we gained from running the assembly program for 200 loops, the CPU elapse time for running the program on computer C was 18487 ms, while on computer B was 18861 ms, and on computer A was 18882 ms. By looking at each CPU elapse time for each computer, we can prove that computer C has a better performance compared to computer B and Computer A.

Next, based on the information that we gained from running the assembly program for 500 loops, the CPU elapse time for running the program on computer C was 46734 ms, while on computer B was 47232 ms, and on computer A was 48325 ms. By looking at each CPU elapse time for each computer, we can prove that computer C has a better performance compared to computer B and Computer A.

Next, based on the information that we gained from running the assembly program for 700 loops, the CPU elapse time for running the program on computer C was 65457 ms, while on computer B was 66046 ms, and on computer A was 65497 ms. By looking at each CPU elapse time for each computer, we can prove that computer C has a better performance compared to computer B and Computer A.

Lastly, based on the information that we gained from running the assembly program for 1000 loops, the CPU elapse time for running the program on computer C was 92997 ms, while on computer B was 94627 ms, and on computer A was 94209 ms. By looking at each CPU elapse time for each computer, we can prove that computer C has a better performance compared to computer B and Computer A.

By analysing the CPU running times of three computers, the CPU running time of computer C is most likely shorter compared to computer A and B in every loop. Computer C may have more advanced hardware specification and optimized system configuration, thus leading to faster program execution. This analysis shown that computer C has more powerful processor, higher clock speed, larger cache size, and superior memory management.

Computer A has a longer CPU running time than computer C and B. This indicate that computer A might have a low processing capacity or less optimized system configuration, resulting in a slow program execution. The slower CPU elapsed time could be attributed to hardware limitations such as a less powerful processor, lower clock speed, smaller cache size, or suboptimal memory management. These factors may hinder the overall performance of the computer. If computer A had additional background processes or tasks running at the same time, it could consume CPU resources, resulting in an increase in CPU running time. This could indicate that multitasking ability is limited, or system resources are insufficient.

Table 4.5.1 is the result of performance difference of executing different no of loop in computer A.

Computer A			
No of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (different)
50	35458932	35463855	4923
100	35501875	35511560	9685
200	35547670	35566552	18882
500	35599290	35647615	48325
700	35675905	35741402	65497
1000	35794654	35888863	94209

Table 4.5.1

Table 4.5.2 is the result of performance difference of executing different no of loop in computer B.

Computer B			
No of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (different)
50	84590362	84595107	4745
100	84630133	84639610	9477
200	84658764	84677625	18861
500	84702606	84749838	47232
700	84779220	84845266	66046
1000	84871966	84966593	94627

Table 4.5.2

Table 4.5.3 is the result of performance difference of executing different no of loop in computer C.

Computer C			
No of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (different)
50	83872928	83877561	4633
100	83984810	83994050	9240
200	84071753	84090457	18487
500	84686745	84733479	46734
700	84801888	84867345	65457
1000	85082547	85175544	92997

Table 4.5.3

5.0 Conclusion and Reflection

Lastly, after conducting the CPU benchmarking on three different computers, and executed the assembly program, we would conclude our project, and make some reflection on it. We also would mentioned the reference that we have been referring too throughout this project.

5.1 Conclusion

The analysis of CPU benchmarking on Computer A, Computer B, and Computer C for Project Part 1 shows that there are notable differences that can be seen in terms of their performance. The analysis underscores the importance of hardware specifications, such as processor performance, clock speed, and cache size, in determining overall computer performance, Computer A emerged as the top performer across all benchmarking tools, making it suitable for high-performance tasks and demanding applications. Computer B offers a good balance for everyday tasks and multitasking, while Computer C, although sufficient for basic usage, lags in terms of technology and performance.

Our analysis on Project Part 2 shows that Computer C consistently performed the best in running the assembly program, with the shortest CPU elapsed times across all tests. This suggests that it is well-optimized and efficient, even if it's hardware is not the most advanced. Computer A, despite excelling in multi-core performance and disk benchmarks, had the longest CPU elapsed times, indicating possible issues with single-threaded tasks or system optimization. Computer B showed balanced performance but also lagged behind Computer C in CPU elapsed times. Overall, Computer C proves to be the most efficient for this specific task due to its effective configuration and resource management.

5.2 Reflection

By conducting the COA project on CPU benchmarking and assembly program execution is very useful for us especially who take the course in Computer Science. This is because, this project has made us recognize on computer terminology and extend our knowledge on benchmarking a computer system. Not only that, but we also get to evaluate the performance and some specification of a computer by using a different benchmarking tool.

By using the benchmarking tools such as CPUZ, MiniTools, GeekBench, and Task Manager we have been given the opportunity to look at our own hardware specifications of computer. In the test benchmarking in each tool, we would collect all the detailed information such as type of processor, clock speed, cache size, memory configuration, and other related hardware. In addition, we also must understand the results that we gained, by following the project requirements for each tool.

Next, is the performance evaluation based on numbers of loop and computer. After running the complete codes on three different computers, we would evaluate the time for the codes to be fully run and complete. The CPU elapsed times becomes our main goals, as it would be measured for the efficiency and speed of the program to be executed.

In summary, this project has provided us with comprehensive assessment of our own computer performance. This analysis has made us identify and decide which computer has a very good performance by evaluating based on the specific requirements of our task, considering other factors too such as hardware specification, system configuration, and application performance.

Last but not least, we would like to extend our heartfelt appreciation to our lecturer, Dr. Foad because he has been guiding us with his precious knowledge, expertise, and support throughout this project. Dr. Foad dedication and passion to this subject has motivate us to keep on going and never give up when we feel like we want because he always there to support us and never stop in giving a hand until this project successful. Therefore, thank you again Dr. Foad for your support, guidance, and care.

Reference

<https://www.oit.va.gov/Services/TRM/ToolPage.aspx?tid=10922>

<https://versus.com/en/glossary/geekbench>

<https://www.geekbench.com/>

<https://www.techtarget.com/searchenterprisedesktop/definition/Microsoft-Windows-Task-Manager>

assembly - What's the purpose of the LEA instruction? - Stack Overflow

Appendix

Work distribution

Intro
- Problem background - dania
- Objective - yasmin
- Aims - dania
- Scope anis
Flow chart- yasmin

Benchmarking - ALL
Analysis & comparison - ALL
1) CPUZ - dania
2) Geekbench - dania
3) MiniTools - anis
4) Task manager - yasmin
5) assembly code - yasmin

Visual aids (Graph/chart)
1) Geekbench - anis
2) MiniTools - anis
3) Task manager - yasmin

Conclusion - dania
Reflection - yasmin
Reference - anis

Video presentation:-

Introduction - dania
Problem background - dania
Aim & objective - yasmin
Scopes - anis
Demo benchmarking CPUZ - dania
Explain analysis - dania
Demo benchmarking Geek bench - dania
Explain analysis - dania
Demo benchmarking Minitools - anis
Explain analysis - anis
Demo benchmarking Task Manager - yasmin
Explain analysis - yasmin
Demo benchmarking assembly code - yasmin
Explain analysis - yasmin
Conclusion - Anis
Reflection - All